

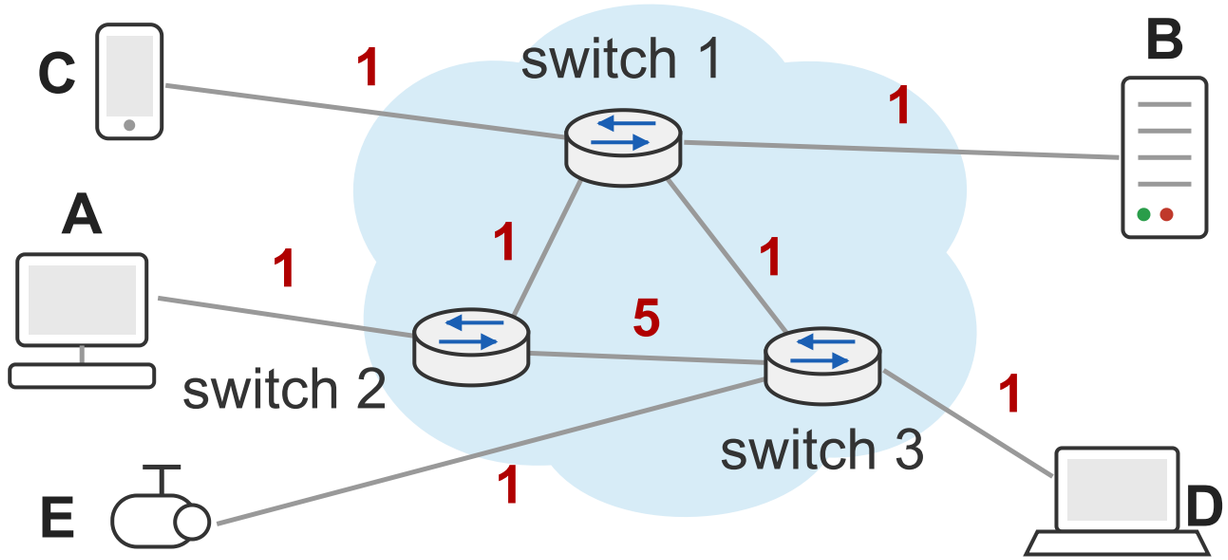
UNIVERSITY OF WATERLOO

Finding paths - Part II

CS 456/656, Fall 2026

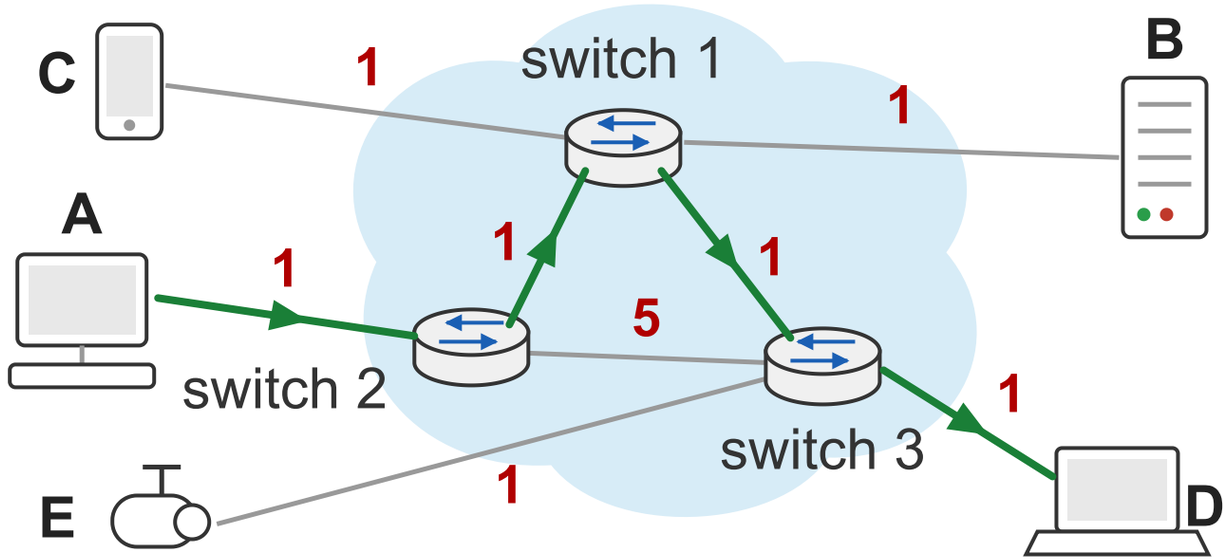
Mina Tahmasbi Arashloo

Recap: least-cost paths



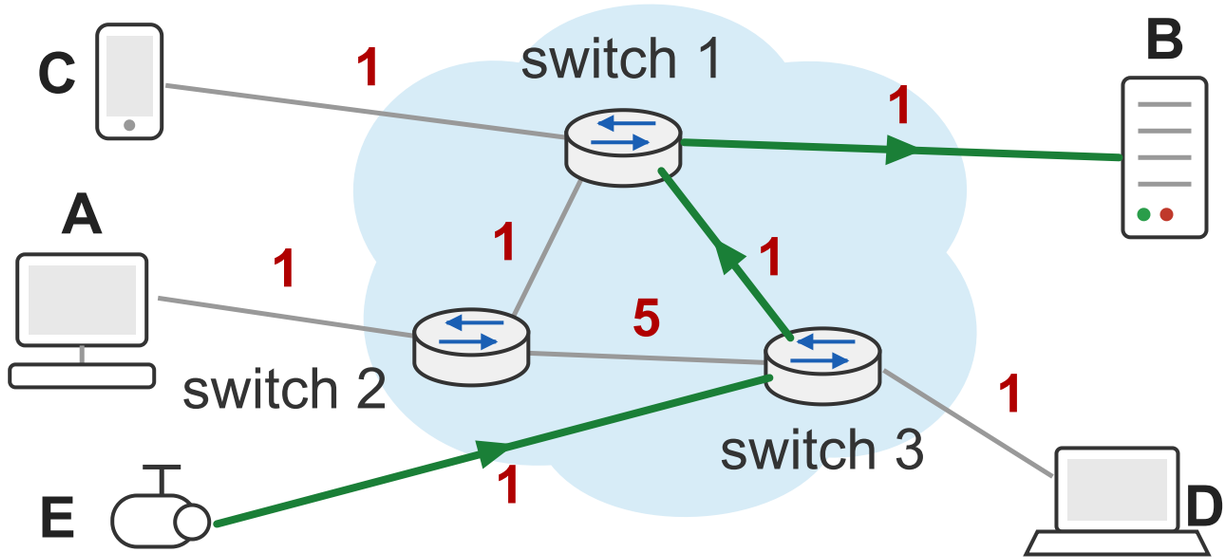
- Every link is given a cost (e.g., by the operator and/or measured by the switches)
- Best path: the one with the least total cost (sum of its links' costs)
- The routing algorithm (distributed or centralized) finds the least-cost paths and populates each forwarding table (destination → next hop)

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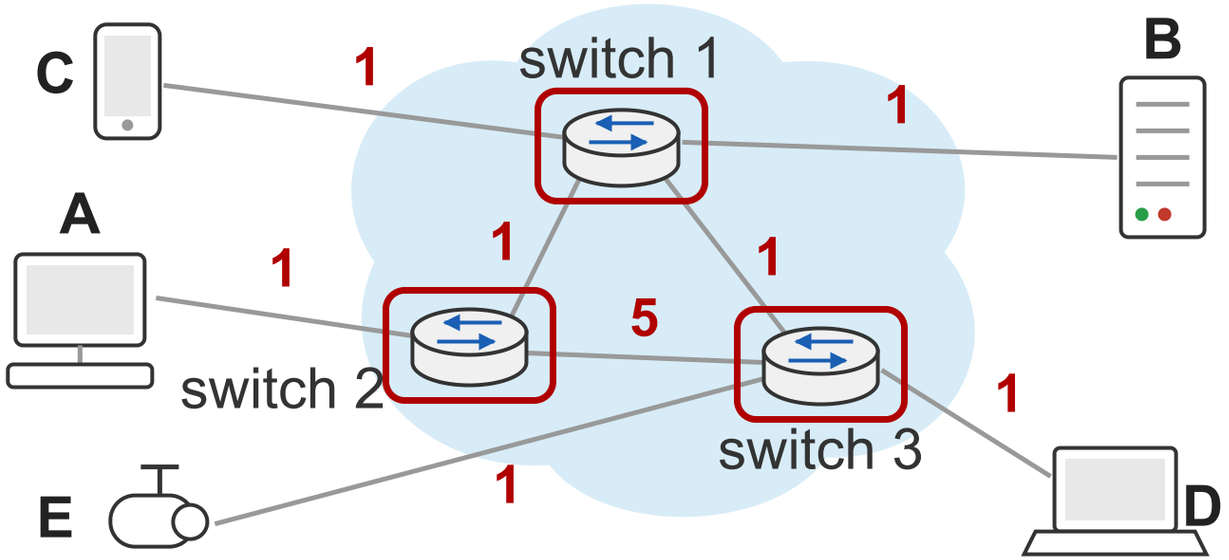
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Implementing least-cost routing in a distributed control plane

- Suppose there is no controller.
- You can write one program that runs, independently, on the control plane of each switch (in red).
- The program on one switch can send and receive messages to and from the program on the neighbouring switches.
- Goal: each switch should learn the next hop on the least-cost path to every destination.



The control plane program (in-class)

```
void on_start() {
    // TODO
}

// runs when msg arrives on link k
void on_message(k, msg) {
    // TODO
}
```

nothing saved yet | load | save | download as .qmd | Reset

Assume no link failure, no change to costs, and no messages are lost.

- `switch_id`: this switch's id, e.g. `1`
- `links[k]`: link k's `.neighbour` (a switch or a host) and `.cost`; k = 0, 1, 2, ...
- `send_msg(k, msg)`: sends `msg` over link k
- `fw_table[dest] = k`: packets for `dest` go out on link k

Routing algorithm 1

Routing algorithm 2

Reacting to link changes (in-class) (in-class discussion)

- How would you handle a change to link availability or cost in routing algorithm 1?

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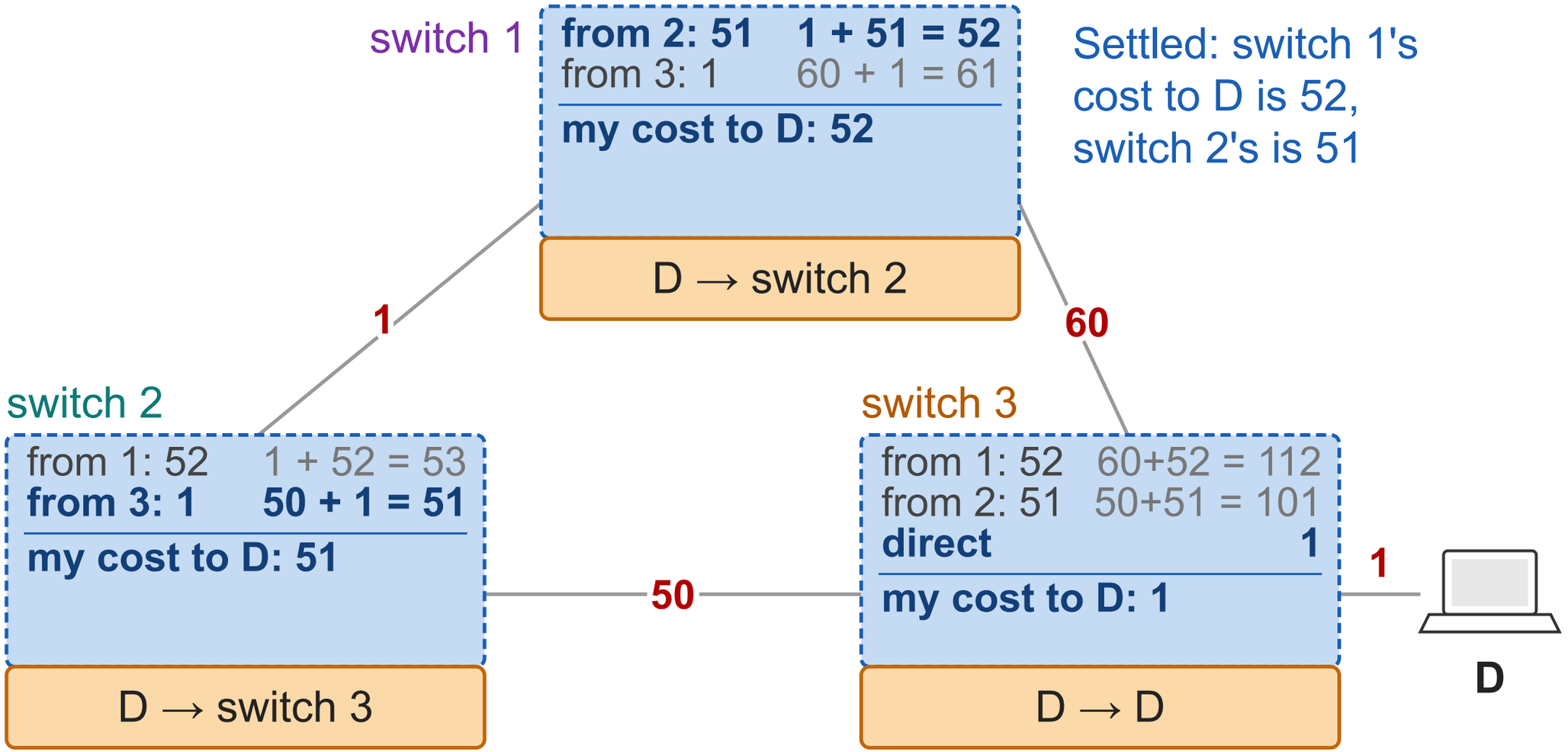
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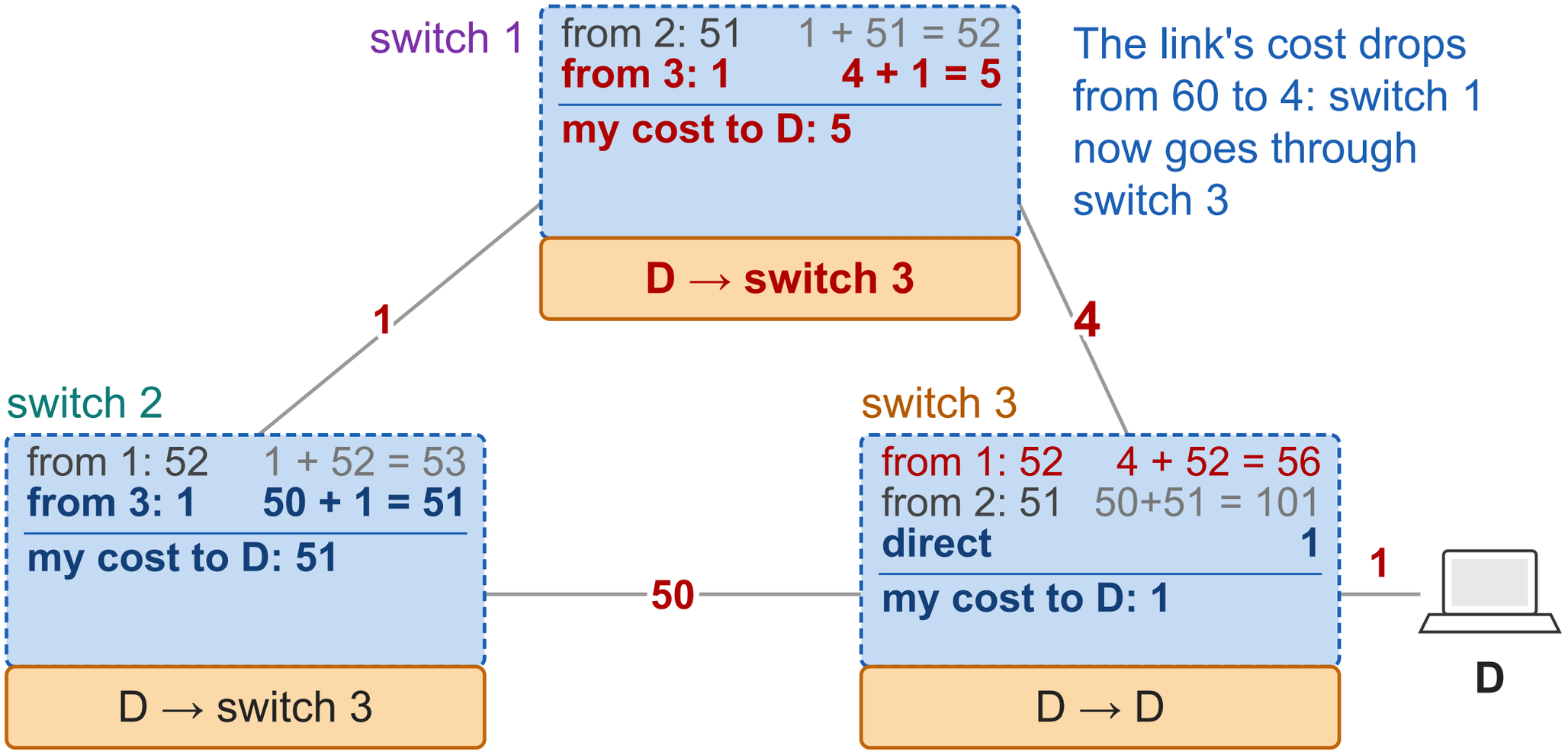
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- What about routing algorithm 2?

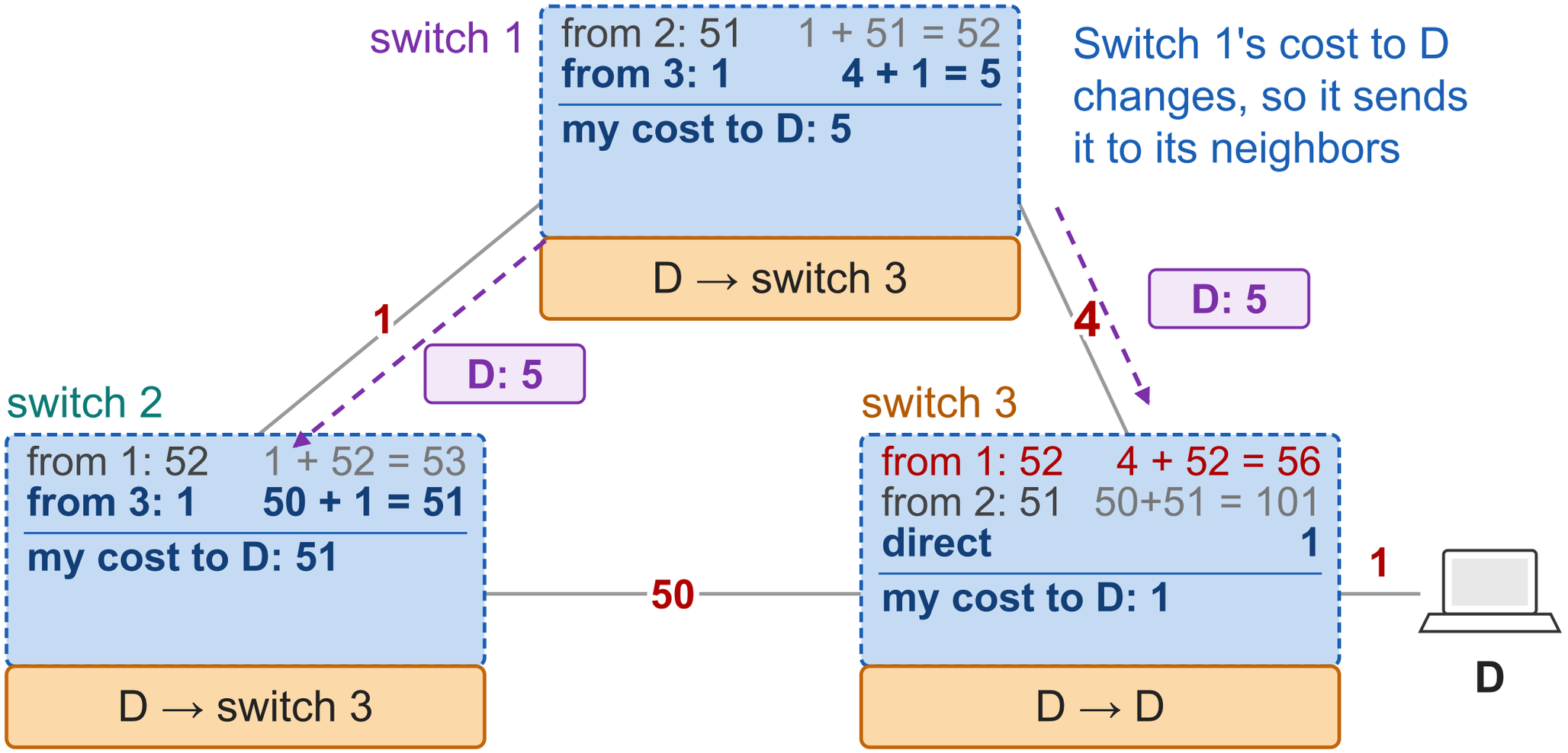
Routing algorithm 2: Link cost decreases



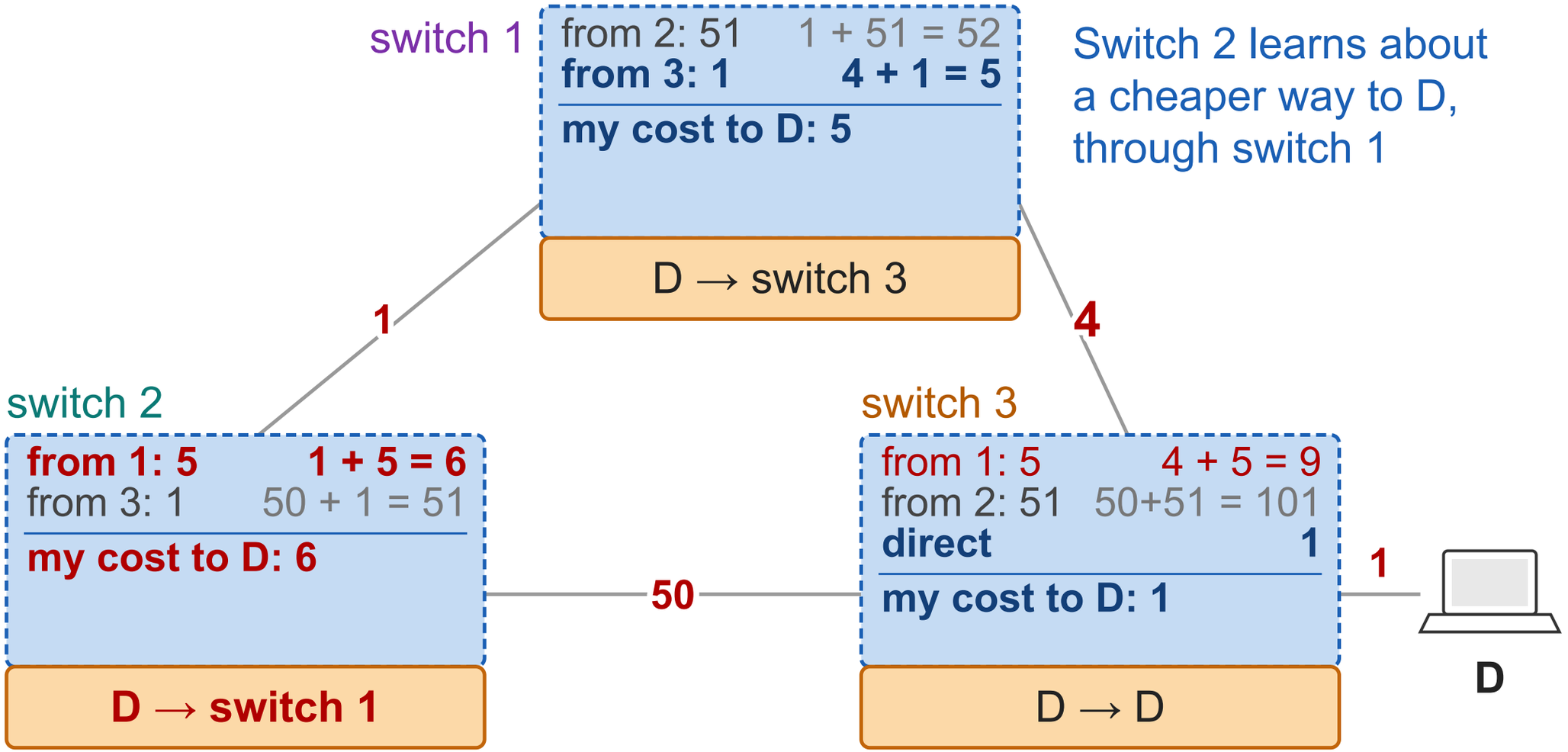
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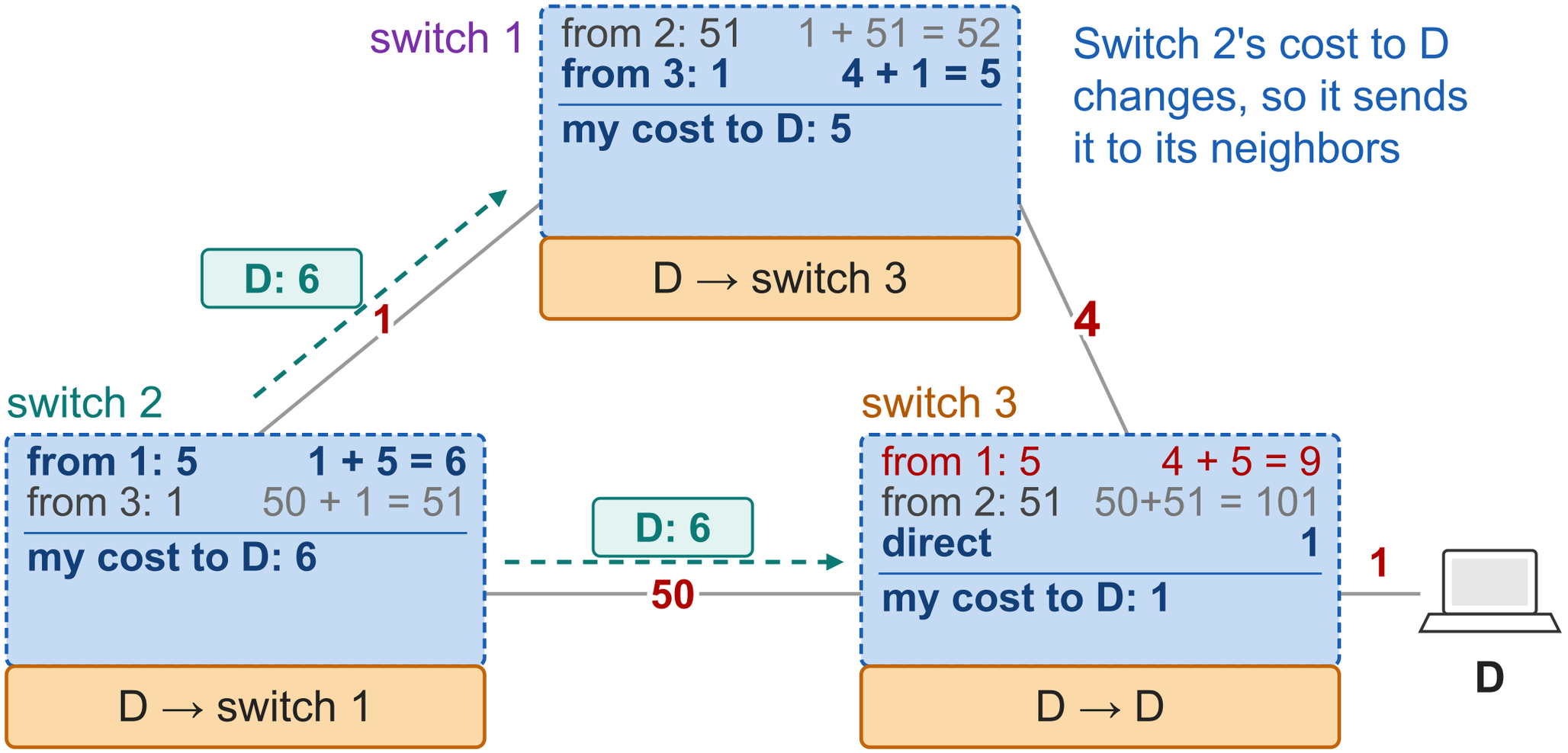
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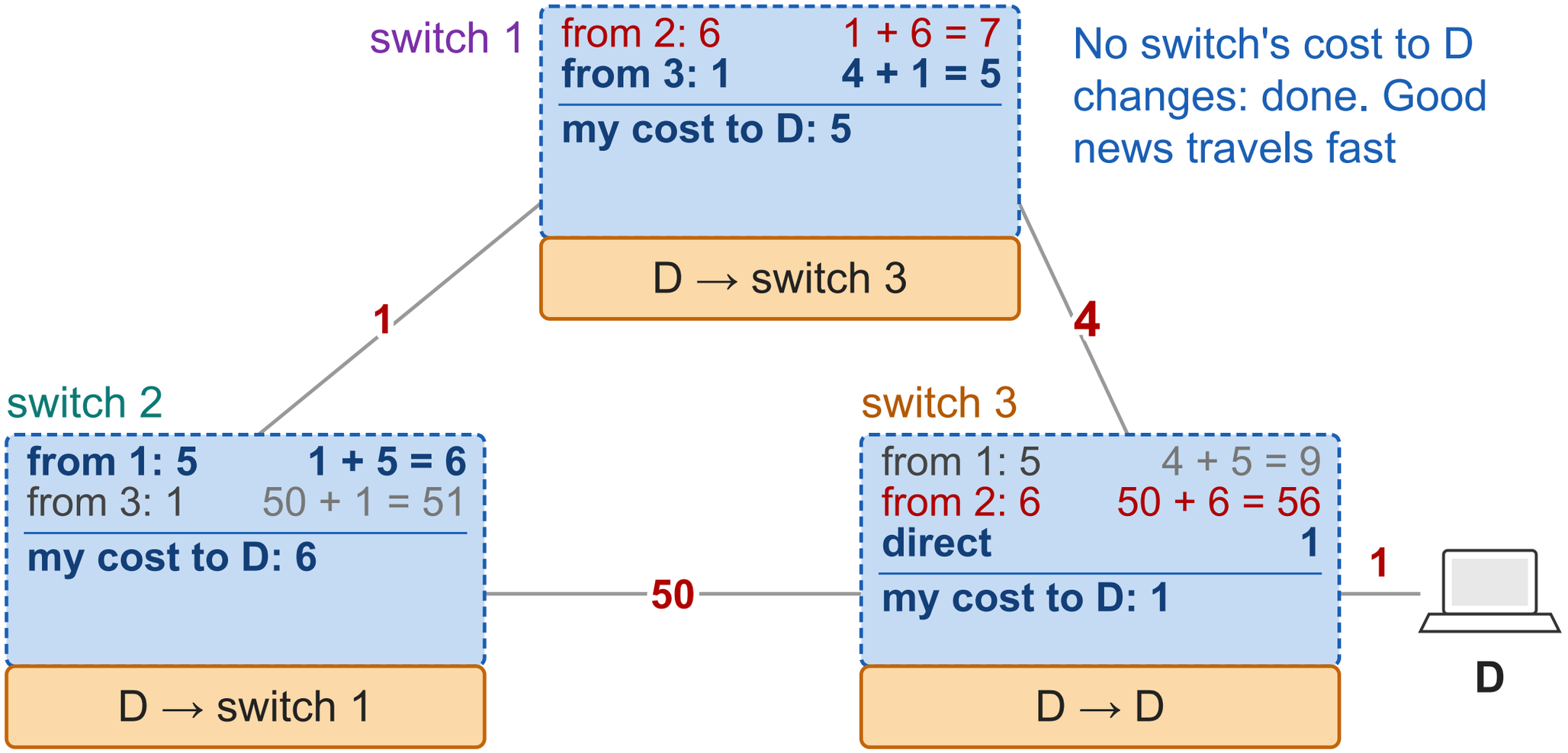
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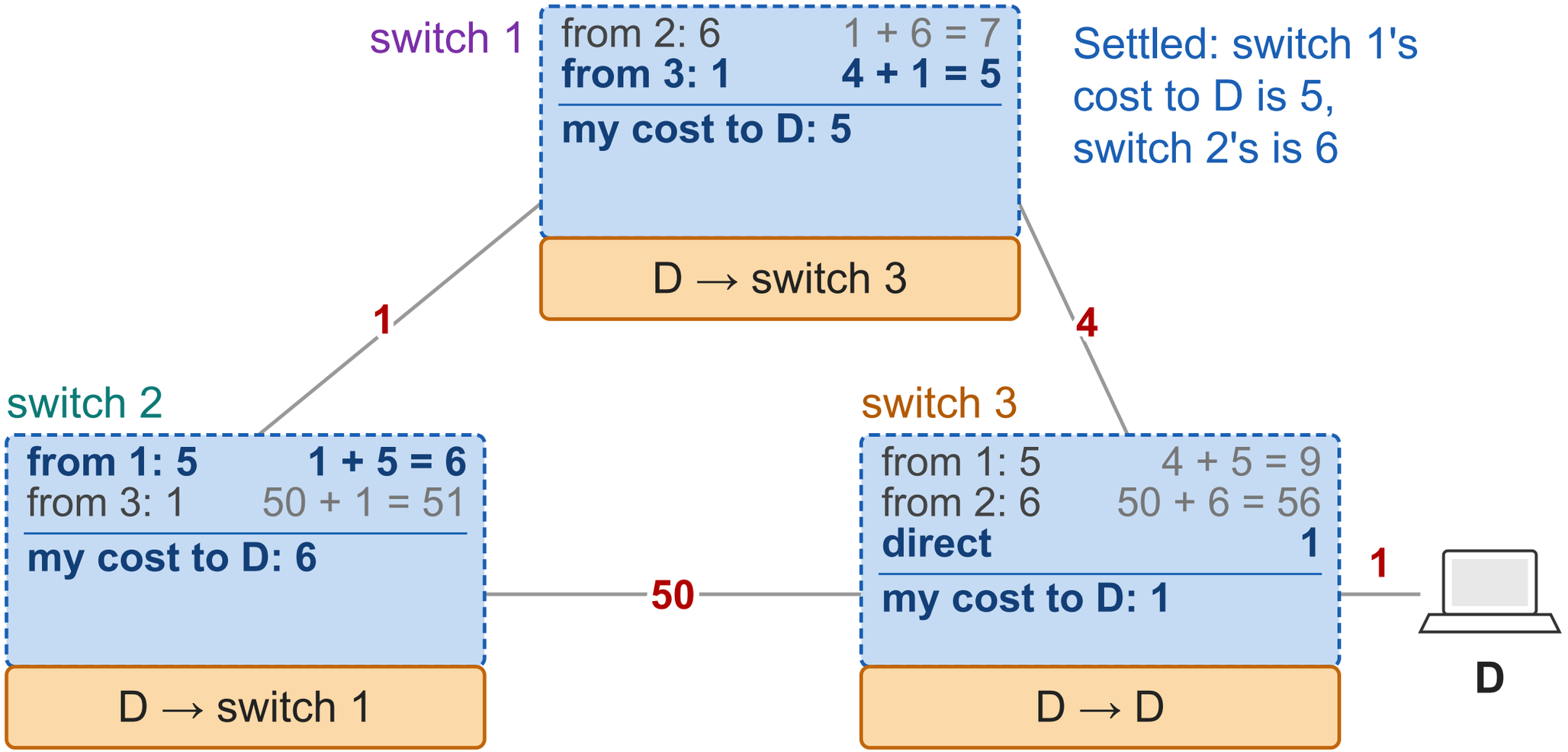
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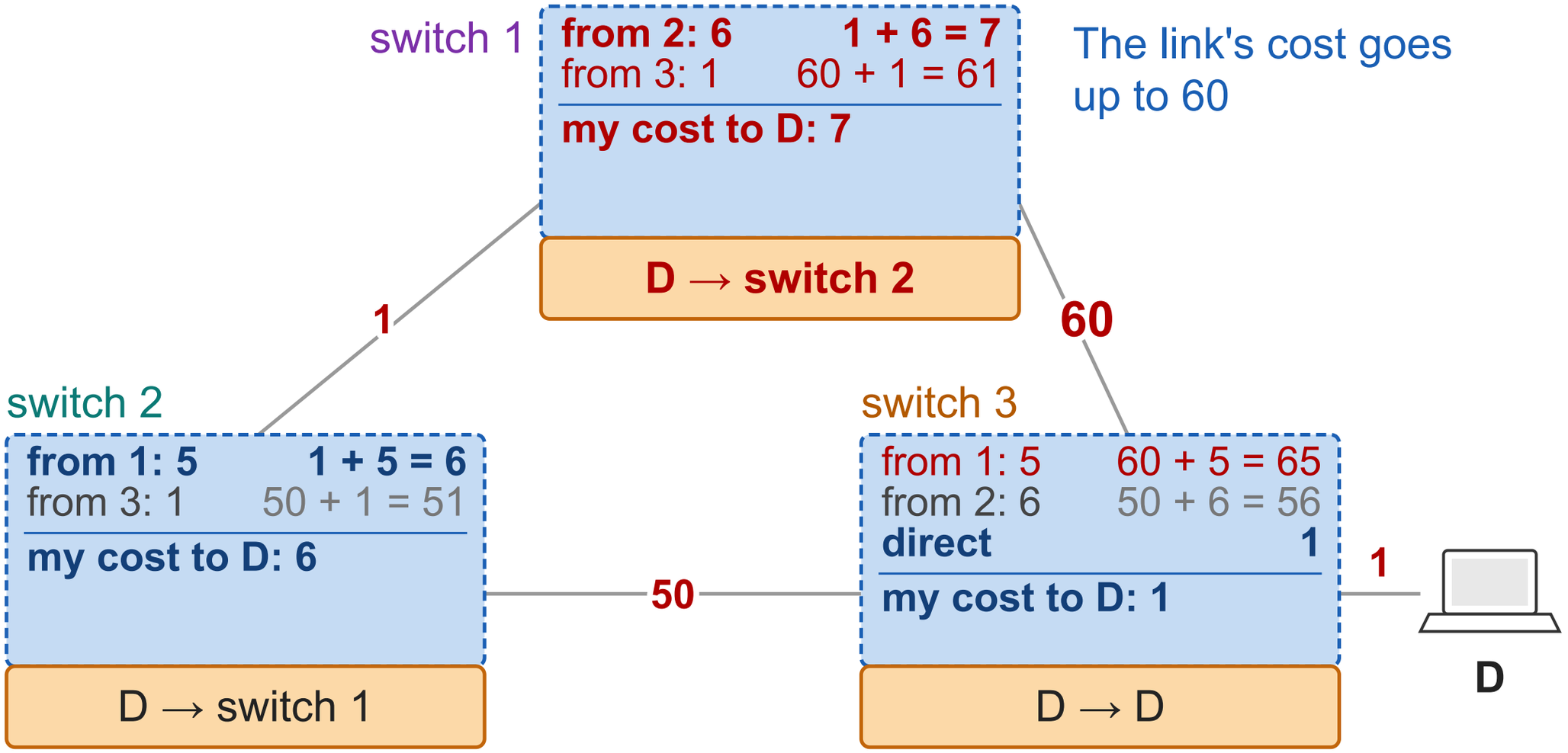
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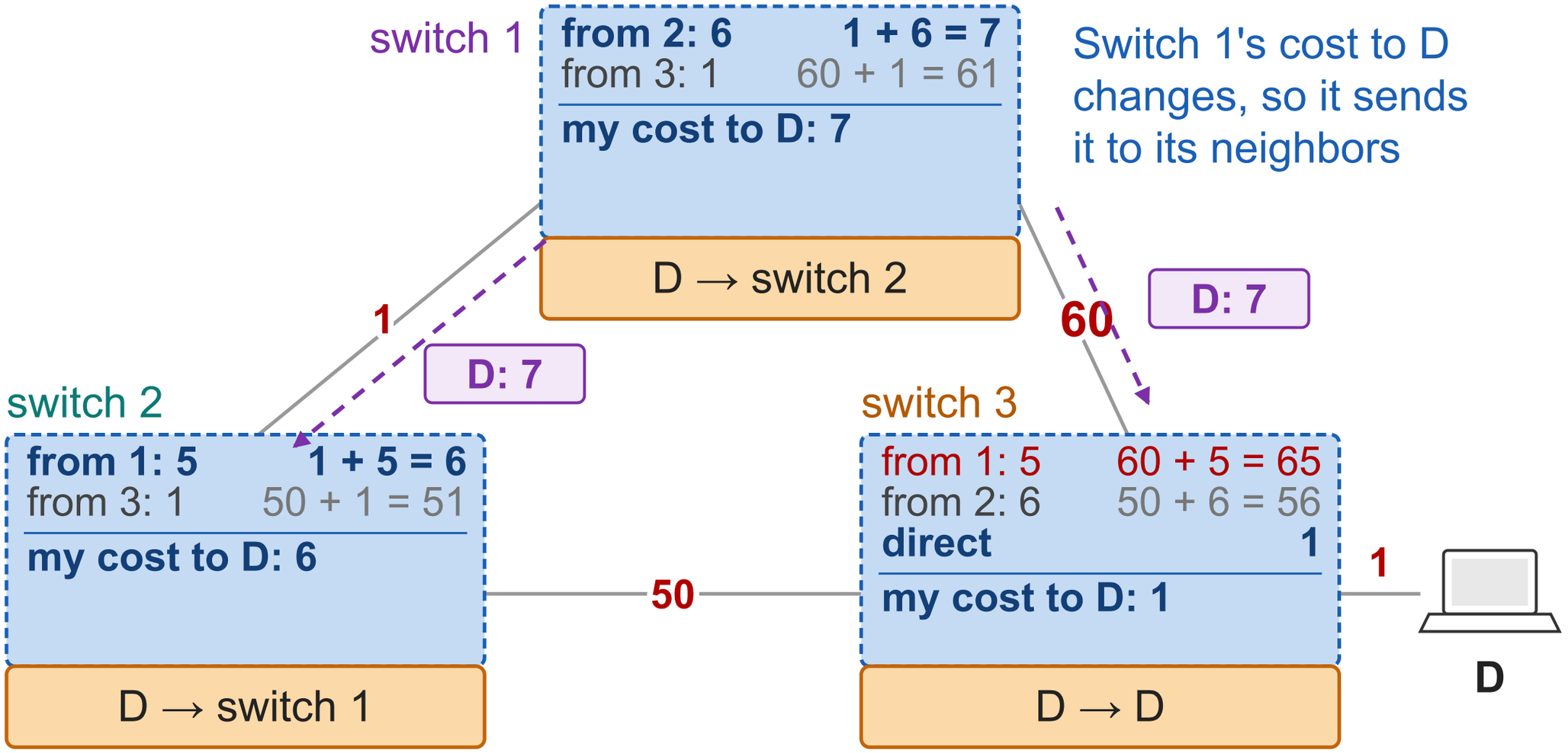
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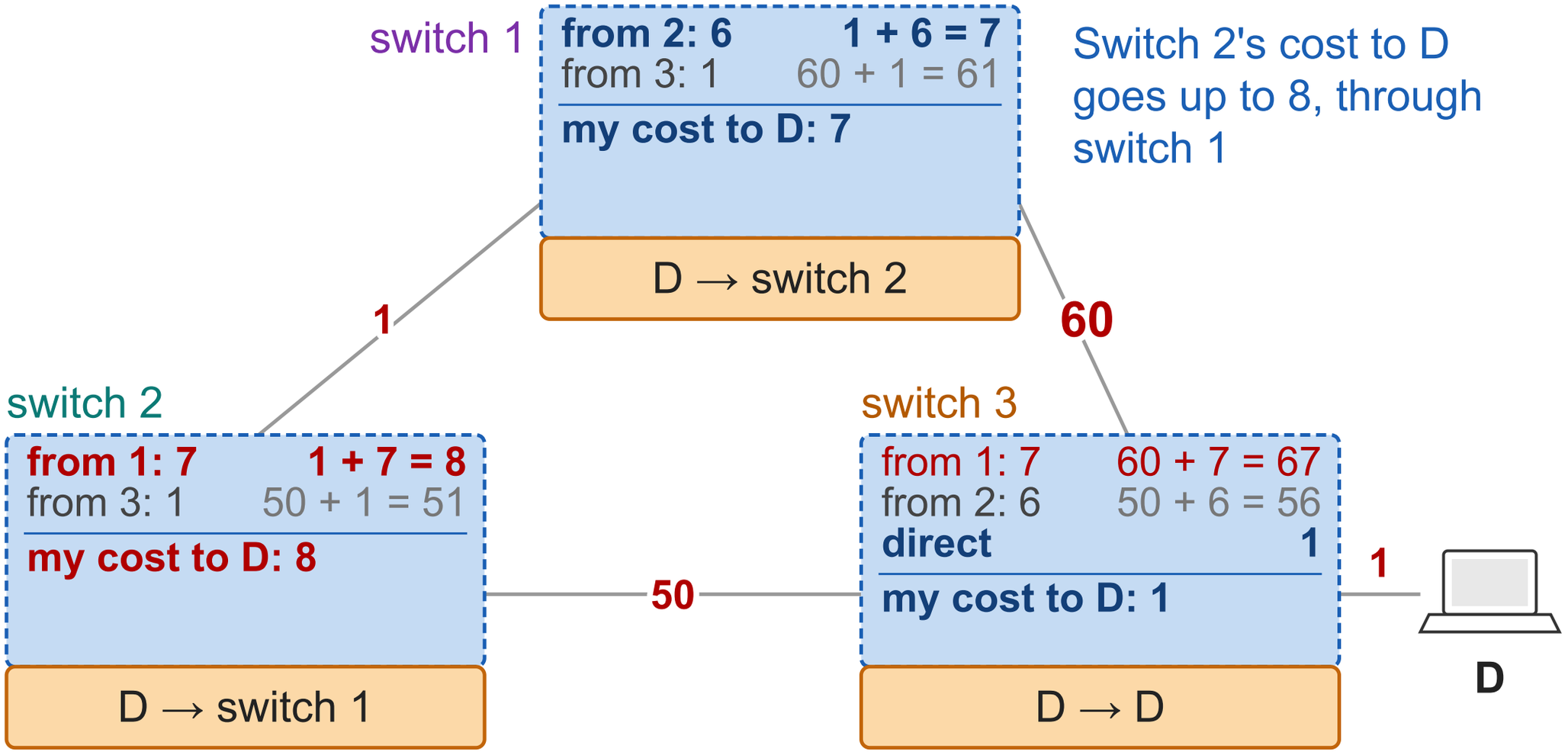
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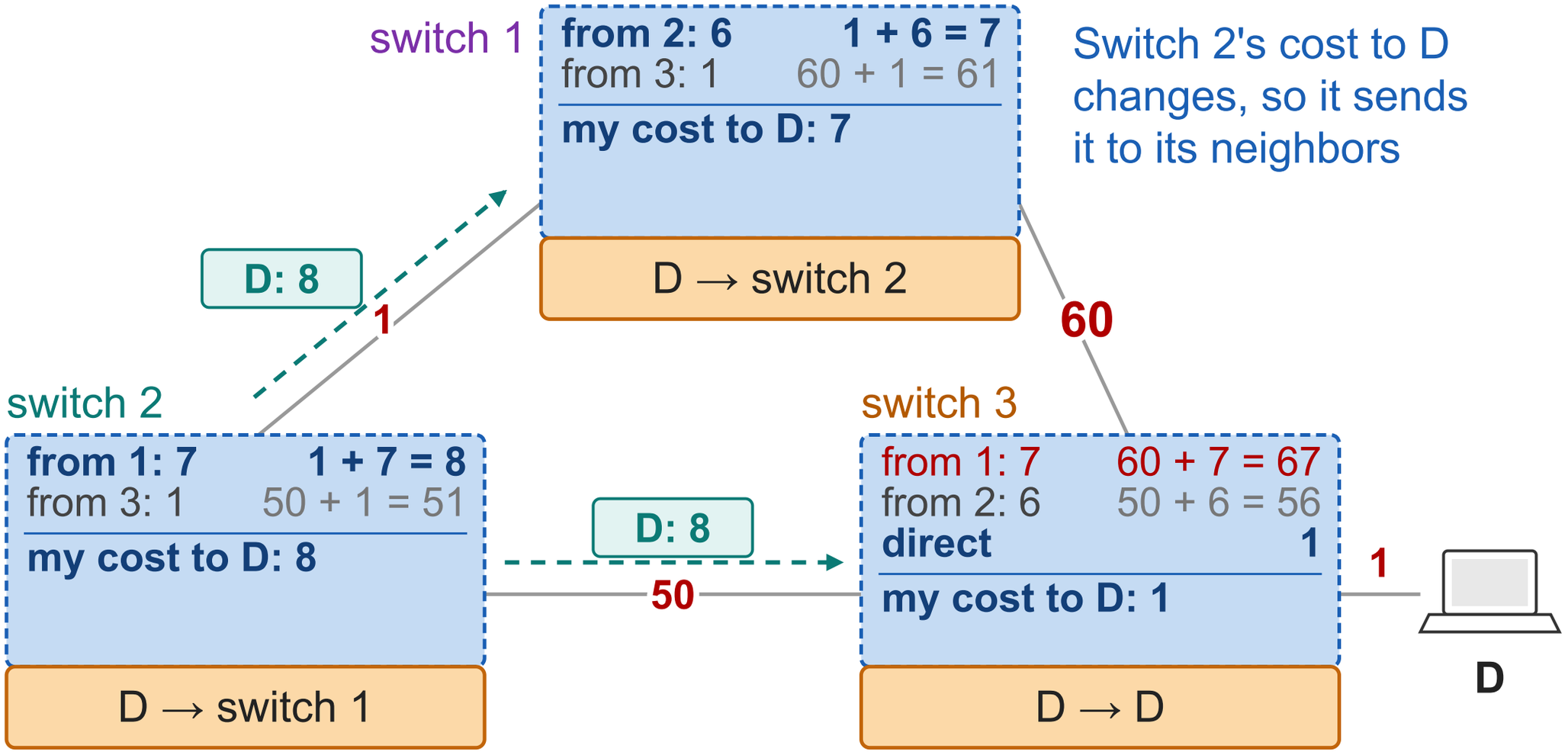
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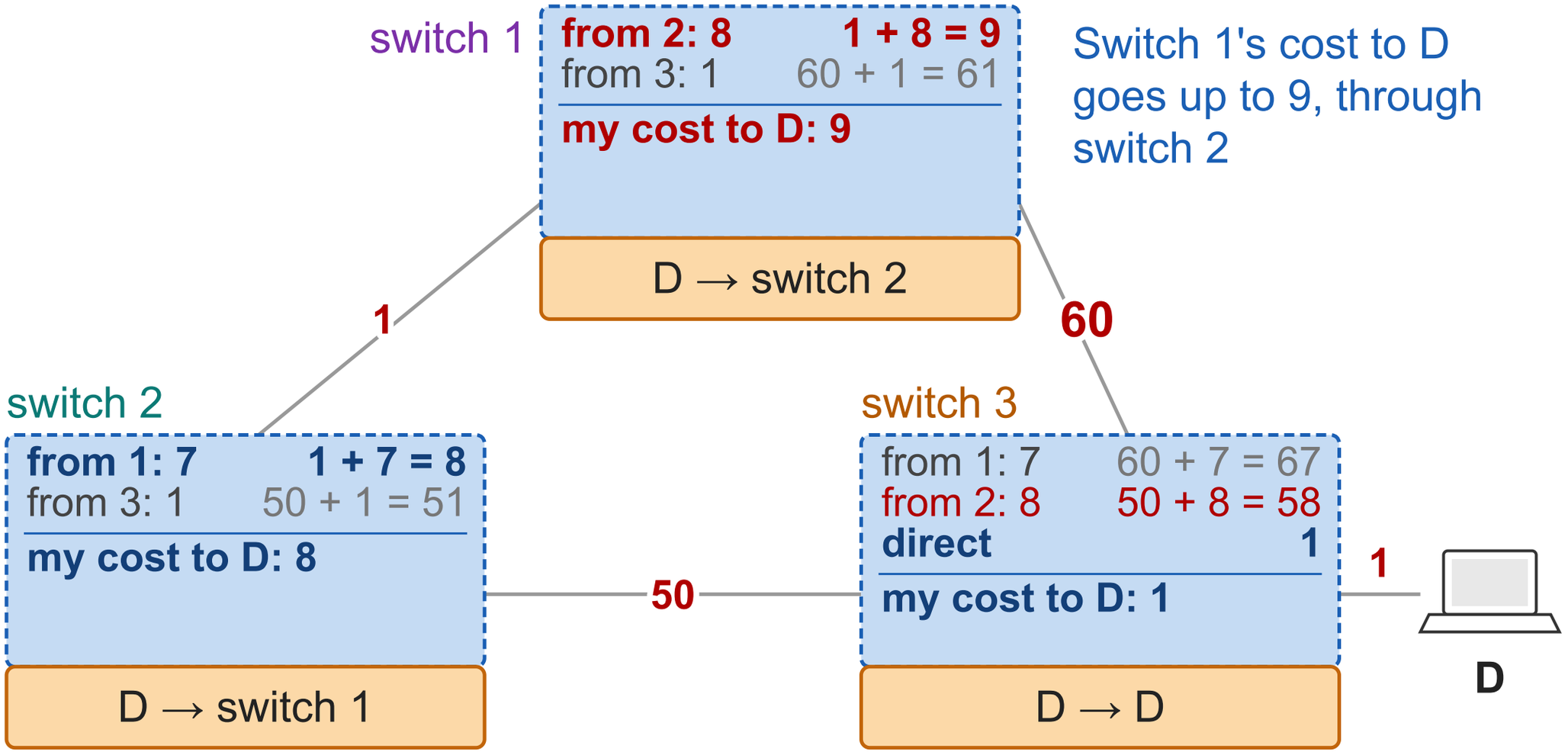
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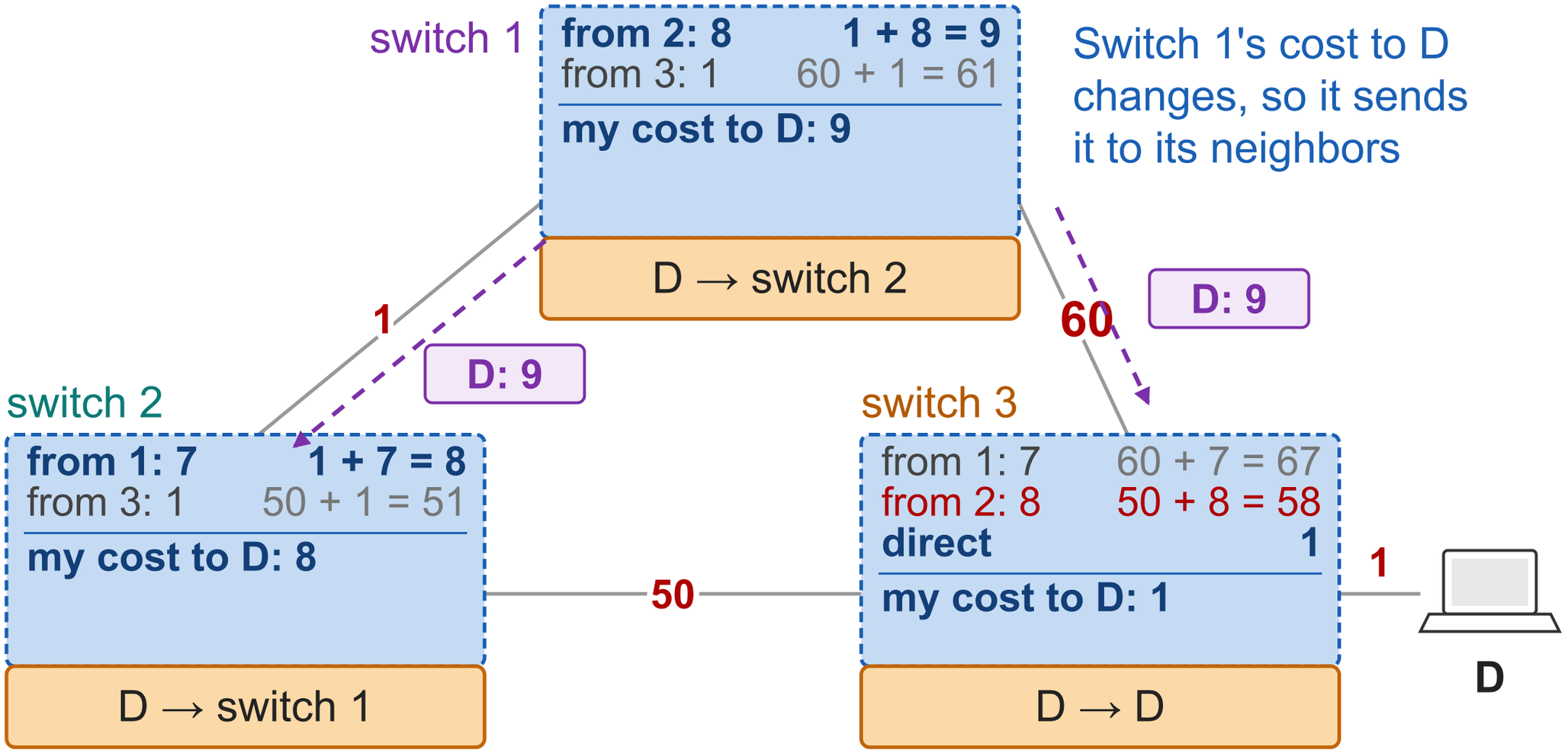
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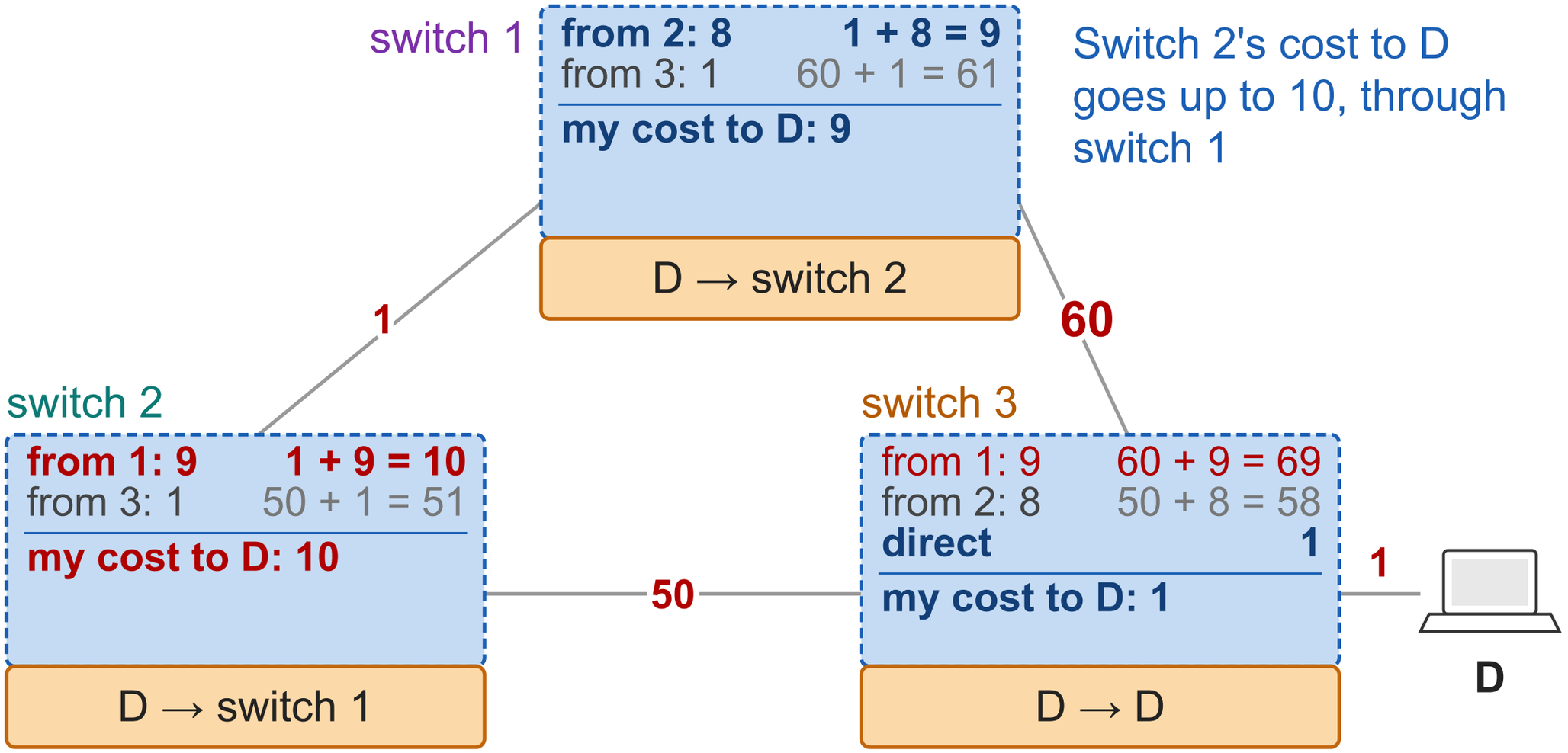
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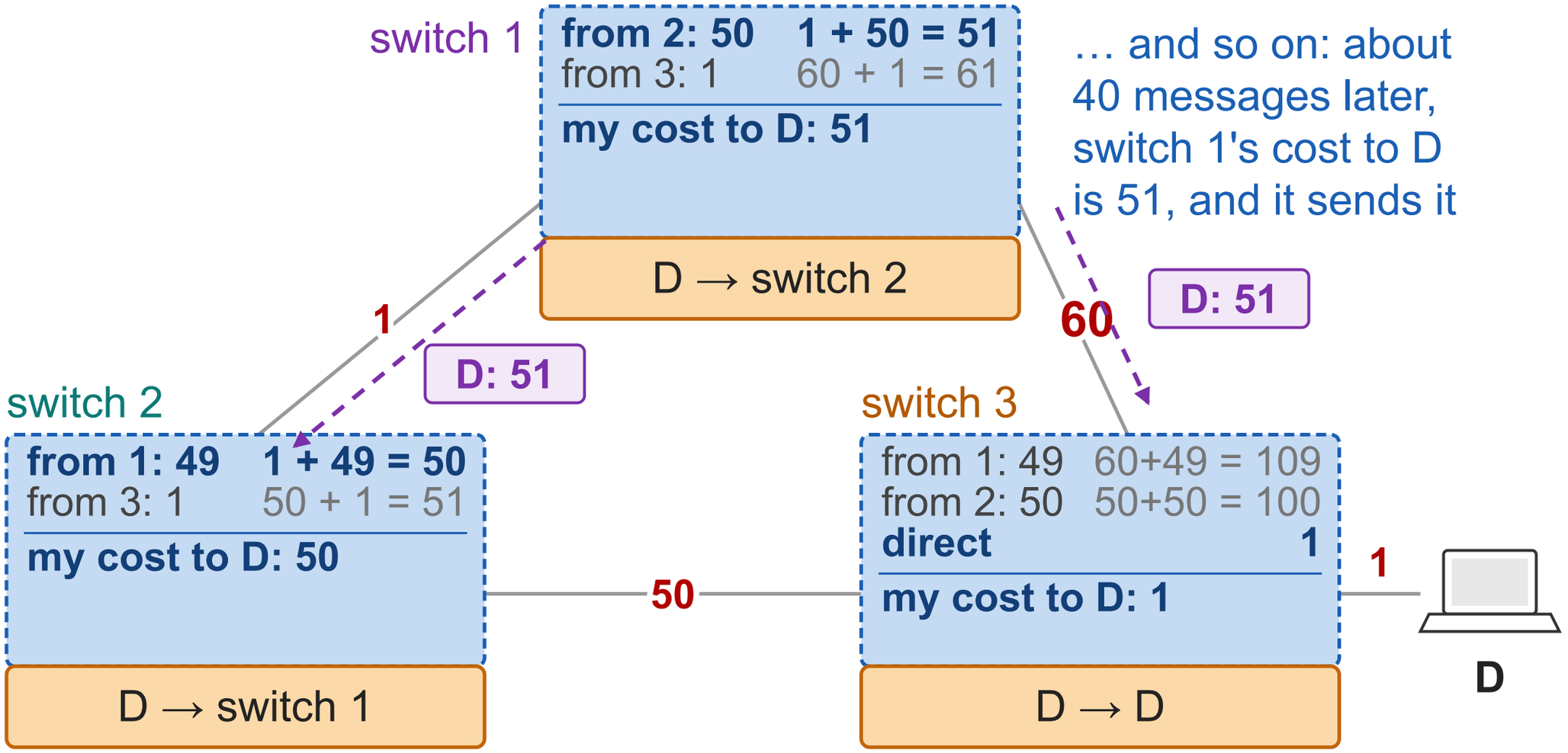
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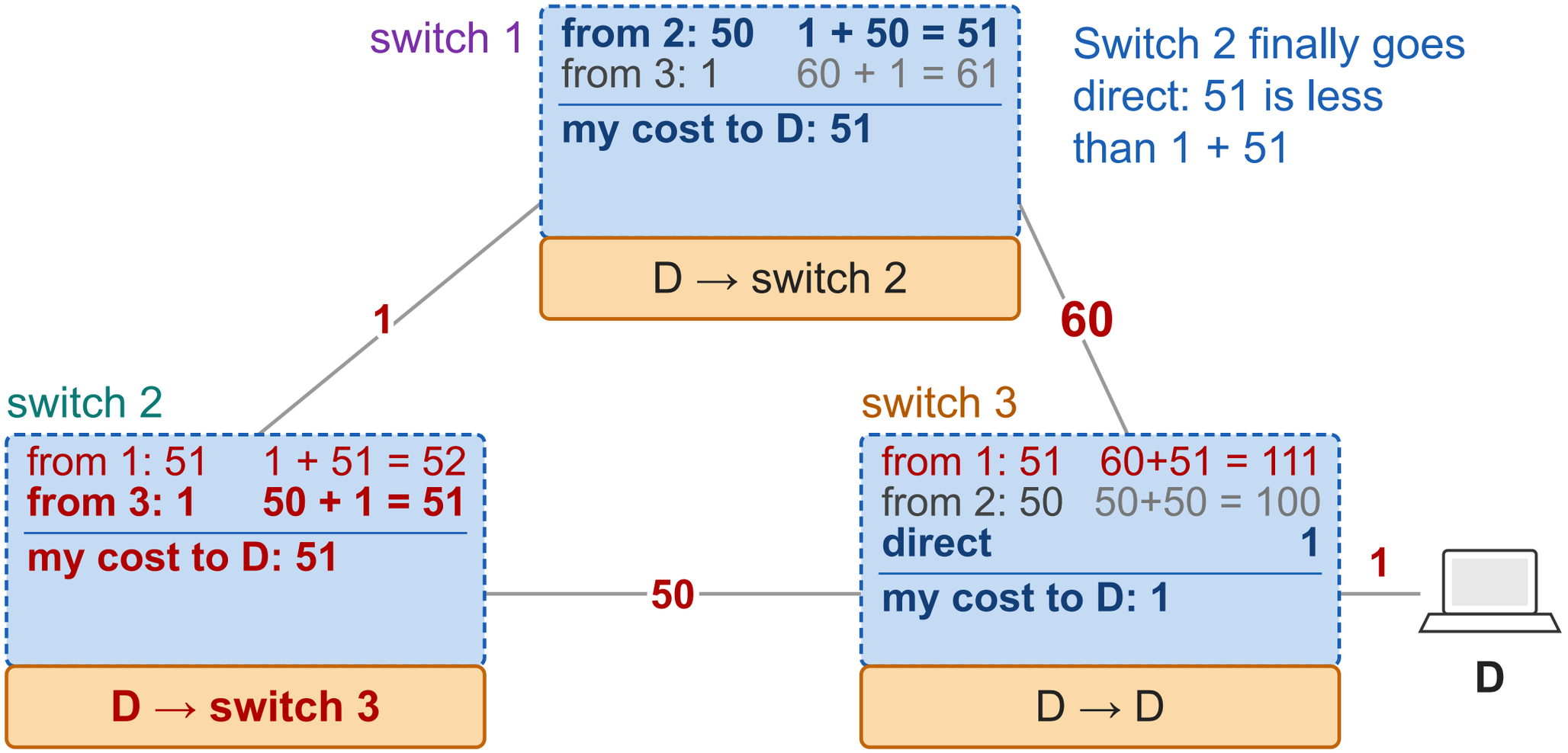
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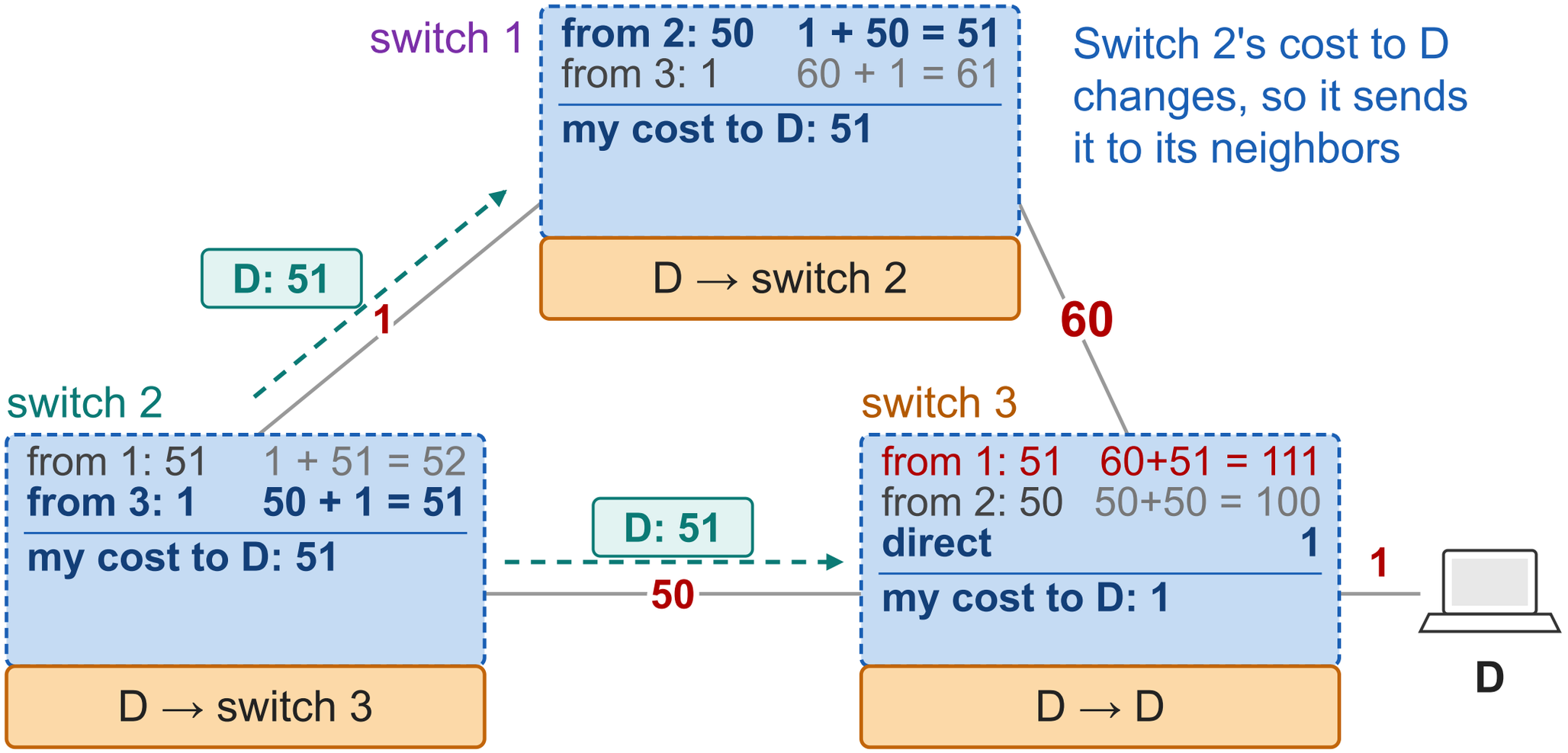
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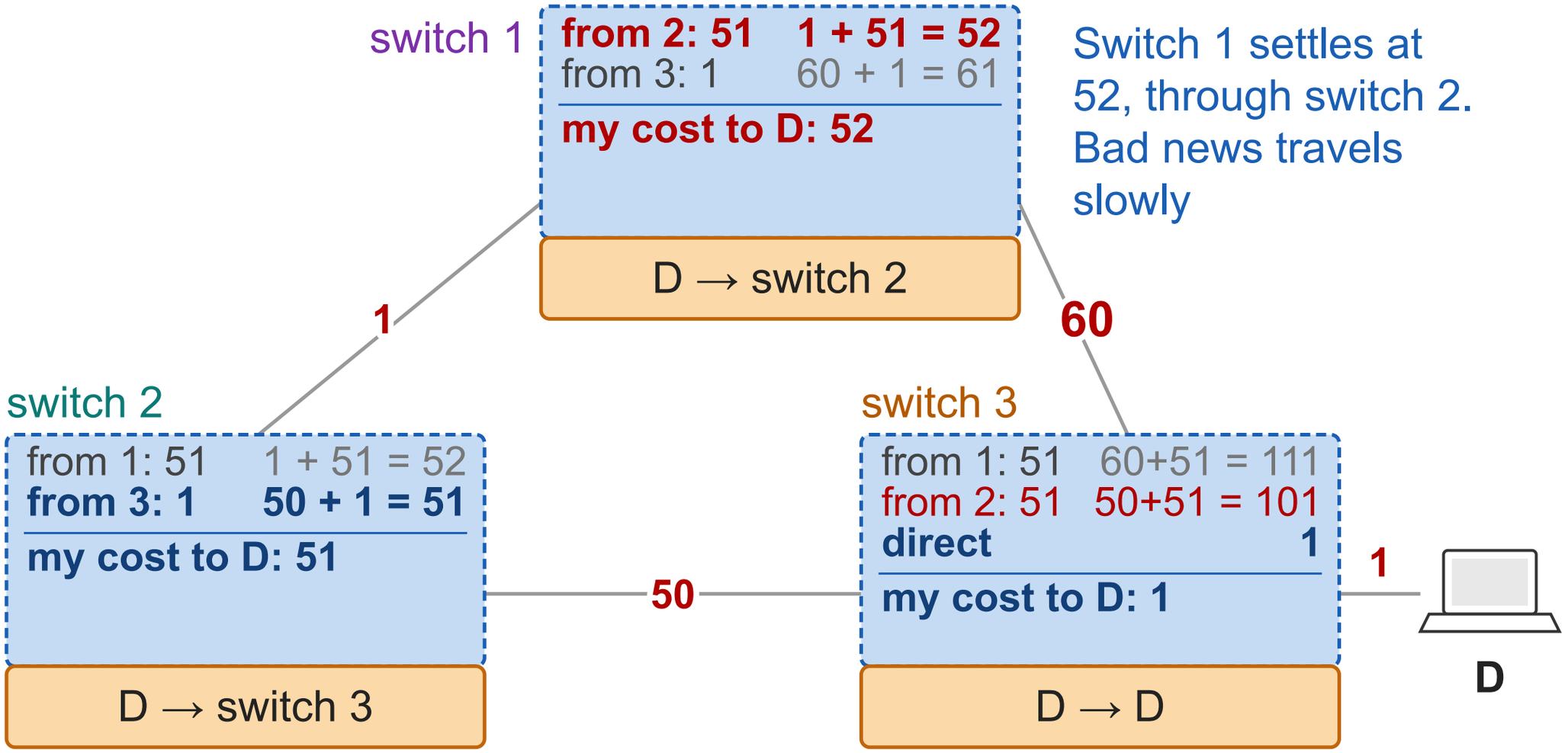
Routing algorithm 2: Link cost increases



Routing algorithm 2: Link cost increases



Routing algorithm 2: Link cost increases



What went wrong? (in-class) (in-class discussion)

- What is the problem? What happens to packets for D meanwhile?
- How would you fix it?

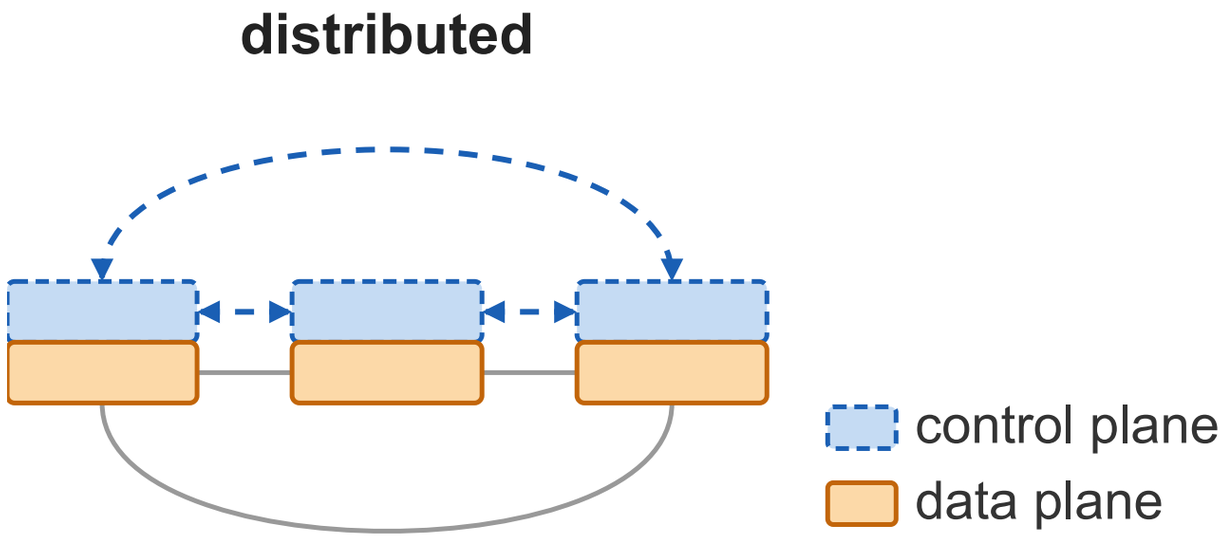
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Comparing routing algorithms (in-class)

Metric	Routing algorithm 1	Routing algorithm 2

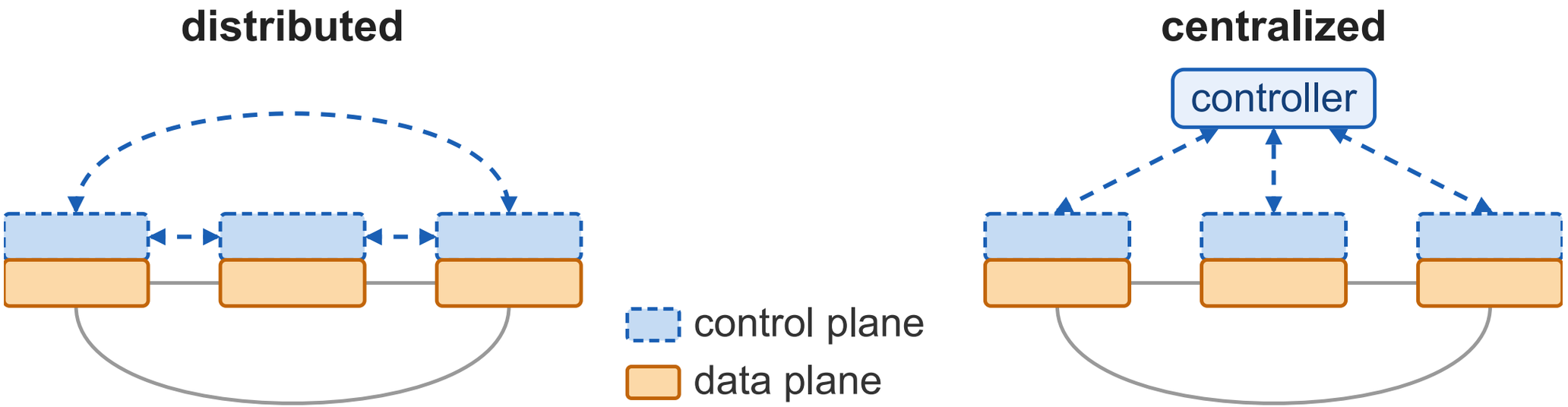
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Who decides the paths?



Switches communicate with each other to find the best paths in a distributed manner.

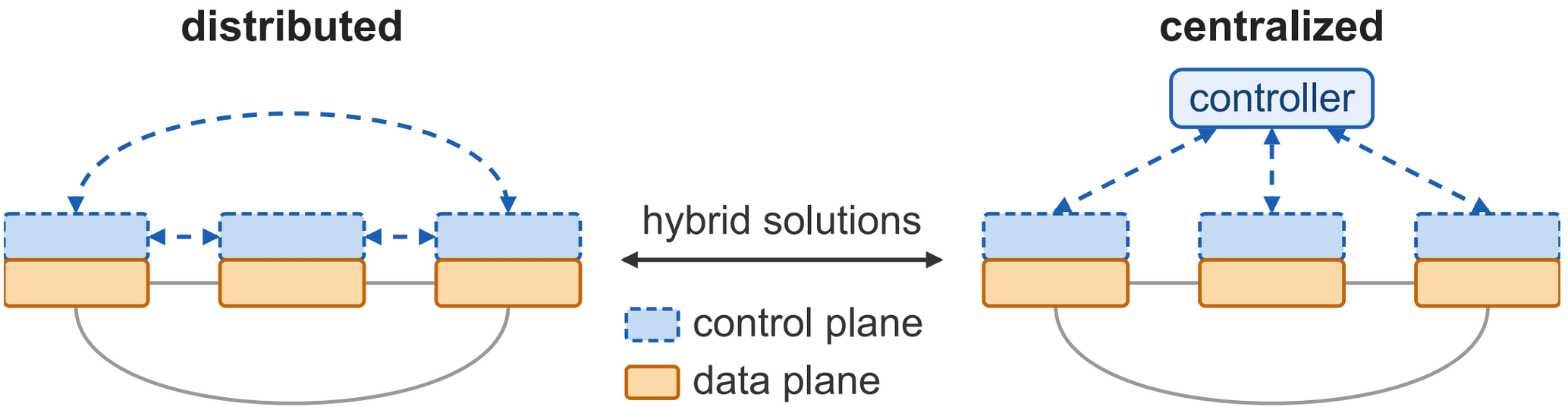
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Centralized vs distributed (in-class) (in-class discussion)

- What are the pros and cons of a centralized and a distributed control plane?

Reset